GEOMETRIC REDUCEDNESS OF A FIBRE OF A SPACE MAP IS INDEPENDENT OF THE CHOSEN FIELD-VALUED POINT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6880-6893 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6895-6901, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Dictionary card enabling the geometric-reducedness locus theorems of this chapter (feeding the smoothness-criteria series).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Properties of Spaces, section points (the equivalence-class definition of |Y|); Spaces over Fields, lemma geometrically-reduced-upstairs (field-extension invariance of geometric reducedness for spaces).

Source reference. Original lemma 0E04: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6901–6901. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03BT: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6901–6901. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $y \in |Y|$. The following are equivalent*

1. *for some morphism $\mathrm{Spec}(k) \to Y$ in the equivalence class of y the algebraic space X_k is geometrically reduced over k,*
2. *for every morphism $\mathrm{Spec}(k) \to Y$ in the equivalence class of y the algebraic space X_k is geometrically reduced over k,*
3. *for every morphism $\mathrm{Spec}(k) \to Y$ in the equivalence class of y the algebraic space X_k is reduced.*

Proof. This follows immediately from Spaces over Fields, Lemma 0E04 and the definition of the equivalence relation defining $|X|$ given in Properties of Spaces, Section 03BT. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.